

## Q:10 2-D Fourier transform , spatial and frequency domains, homomorphic filter

Define 2-D Fourier transform. With an example, describe the correspondence of filtering in spatial and frequency domains. Describe homomorphic filter.

### Definition of 2-D Fourier Transform (2-DFT)

The 2-D Fourier Transform decomposes an image into its constituent frequency components (sine and cosine waves). It transforms a signal from the **Spatial Domain**  $f(x, y)$  to the **Frequency Domain**  $F(u, v)$ .

For a continuous image, it is defined as:

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

### Correspondence of Filtering: Spatial vs. Frequency.

**Spatial Domain:** Filtering is performed by **Convolution** ( $g = f * h$ ).

**Frequency Domain:** Filtering is performed by **Point-wise Multiplication** ( $G = F * H$ ).

#### Example: Low-Pass Filtering (Blurring)

**In Spatial Domain:** slide an averaging mask (3\*3 ones) over the image. This requires many additions and multiplications per pixel.

**In Frequency Domain:** multiply the image's Fourier Transform by a filter  $H(u, v)$  that is 1 at low frequencies and 0 at high frequencies.

**Correspondence:** Both methods yield the same result (a blurred image), but the frequency domain is often faster for large filters.

## Homomorphic Filter

A Homomorphic Filter is a non-linear frequency domain filter used to improve images with uneven illumination or poor contrast. It separates the **Illumination** (i) and **Reflectance** (r) components of an image.

$$f(x, y) = i(x, y) \cdot r(x, y)$$

**Process:**

**Logarithm:** Takes the ln of the image to turn the product into a sum:  $\ln(f) = \ln(i) + \ln(r)$ .

**FFT:** Converts the log-image to the frequency domain.

**Filtering:** Applies a filter  $H(u, v)$  that attenuates low frequencies (illumination) and amplifies high frequencies (reflectance/details).

**Inverse FFT & Exponential:** Converts back to spatial domain and applies  $e^x$  to restore the image.

**Purpose:** It achieves **dynamic range compression** (balancing light) and **contrast enhancement** (sharpening edges) simultaneously.